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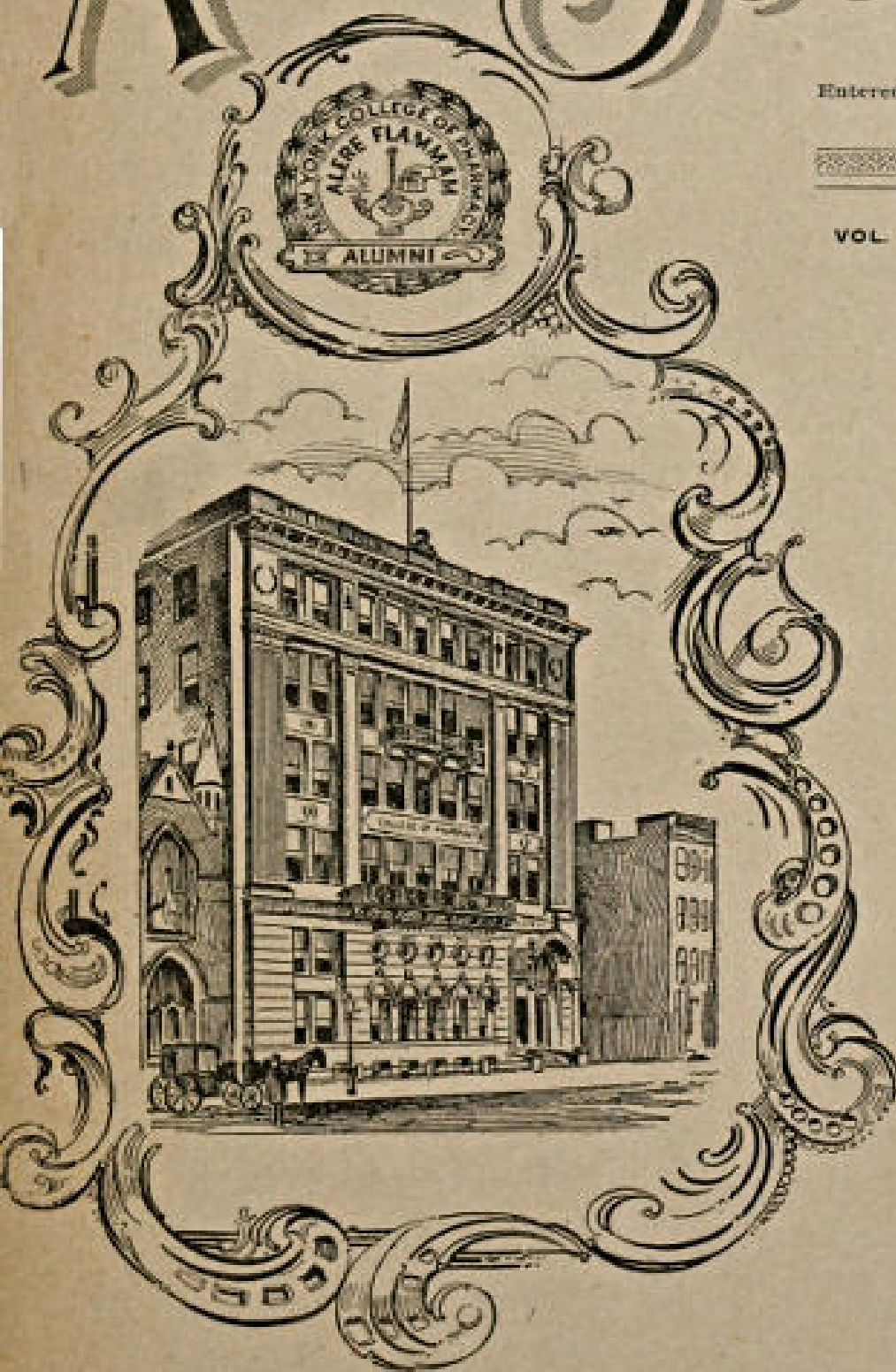
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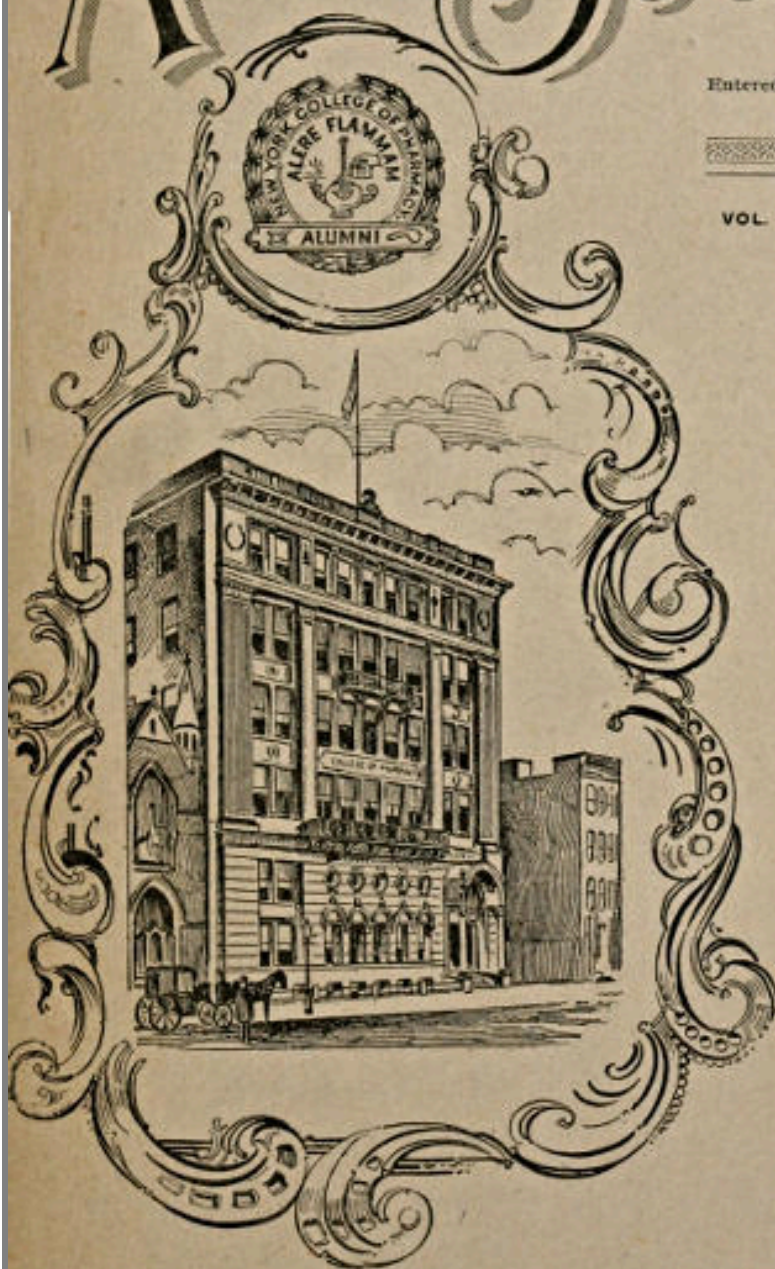
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February, 1895**

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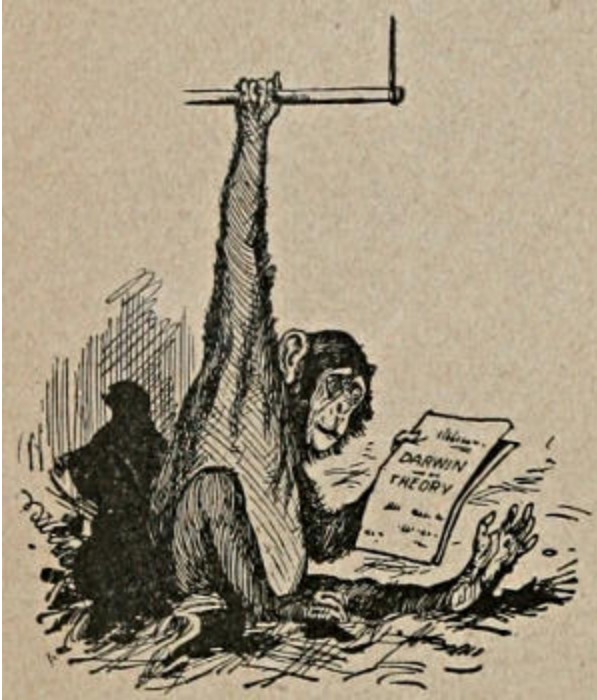
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The
Connecting
Link

between the crisis and the complete recovery from an acute disease, that period known as convalescence, can often be considerably shortened by a judicious attention to the patient's nutrition. The battle has indeed



been won, but the soldier is left prostrate upon the field.

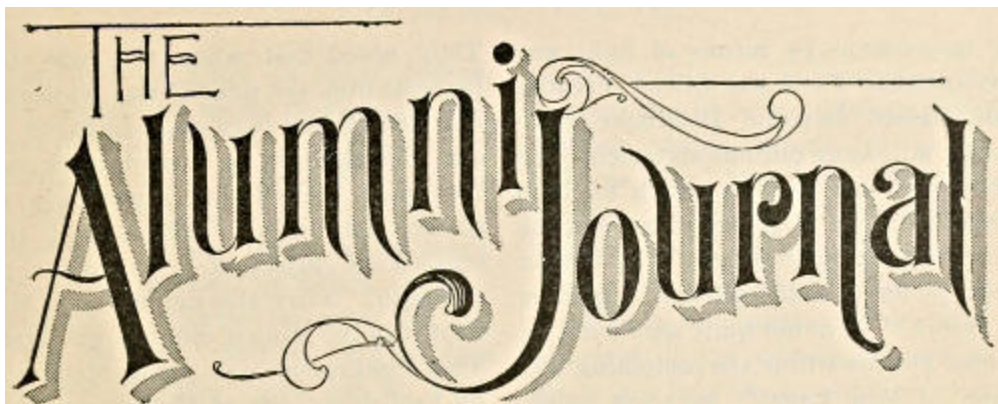
Liquid Peptonoids

provides a valuable auxiliary for his up building because it is a liquid food-agent possessing a powerful reconstructive action while at the same time it is slightly stimulating in its primary effects. It is entirely pre-digested and in an absolutely aseptic condition. In convalescence, Doctor, give your patient LIQUID PEPTONIDS

“That so he might recover what was lost.”

(Henry VI.)

THE ARLINGTON CHEMICAL CO.,
Yonkers, N. Y.



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“THE RISE AND PROGRESS OF PHOTOGRAPHY.”

BY PROF. ARTHUR H. ELLIOTT, PH.D., F.C.S.

The topic of my lecture this evening is one of my old hobbies, so that if I am a little prolix sometimes you must pardon me. It is something in which I have been more or less interested for the last twenty-five years, and, like most of our hobbies, we sometimes drive them to death, to the discomfort of other people.

The fundamental ideas upon which photography is based are very old—older than the Christian era, certainly. They depend upon two facts: First—that light, in passing through a small opening, produces an inverted image in a dark chamber. Imagine, for instance, that you are in a dark chamber, outside of which is an object; that there is in the chamber a small hole a sixteenth or an eighth of an inch in diameter, and that you have in this dark chamber a piece of paper. Upon that paper you will get a picture of the object opposite the hole. That was known a long time ago. The other fact is that certain salts of silver, notably the chloride, iodide and bromide of silver, are sensitive to light and become blackened by light, was known to the Egyptians. The action of light upon colored bodies must have been known to the very earliest observers among men. The bronzing of the human skin under the tropical sun must have been noted by every one; and it is on record, in the most ancient annals of the human race, that men—the fair men from the North—when they went to the tropics, returned with tanned skins. Ptolemy, over two thousand years ago, noted that beeswax was bleached in sunlight, and the old Greeks noted that the gems which we call opal and amethyst lost their colors when exposed to sunshine. These are some of the first and most rudimentary notions upon the actions of light, and we have no definite statements about making pictures without light. The Chinese have a tradition—and they have a great many curious ones that are often founded on facts—that the sun makes pictures upon the ice of lakes and rivers.

A Frenchman, named Fontamen, wrote an imaginary voyage to a strange country, and among other things he said that objects were reflected upon the water and when the water was frozen the images were retained. So this idea of certain surfaces being capable of receiving impressions by means of light was very ancient. There was another Frenchman, named Devique Delaroche, who made a still more curious statement. In 1760 he wrote a book in which his hero is wrecked upon a strange coast, and the spirits of that place showed him how to make pictures, as he called it, "painted by nature." It is not quite sure what he means, but his words are something like these: "You know," says his guide, "that rays of light are reflected from different bodies and form pictures. The spirits have sought to fix these pictures, and have a subtle matter by which these pictures are formed in the twinkling of an eye. They coat canvas with this peculiar matter, and hold it before the object." The manner of holding it is not stated. "The canvas is then removed to a dark place and in an hour the impression is dry and you have a picture, the more precious in that no art can imitate its truthfulness." These words were written one hundred and fifty years ago. This, as far as we know, was purely imagination; yet the idea—the germ of photography—was there. We shall presently see that this flight of fancy on the part of Delaroche was very near the truth, and foretold what has since become possible, and only a very short time after he said it.

As time went on and observations of men became more definite, we obtain records of facts that were noted with regard to the action of light upon certain chemical compounds. You know those old alchemists had queer ideas, one in regard to their elixir of life, and another that they could turn the baser metals into gold. They discovered a material in the silver mines of the Hartz Mountains which they called "luna cornea." The word luna was at that time applied to silver. Luna cornea was horn silver—what we know to-day as silver chloride. They noted that when this was first brought from the mine it was white and that after it had been exposed to the air and the sunlight it turned black, and they also noticed that it was only the surface that turned black—that if they scraped the surface off it was white underneath. They also found that if they kept it in the mine it did not get black. This observation was made about 1550 by Frobrishes, one of the early workers in chemistry; but you must remember that they were not studying the action of light upon this substance. Their sole object was the

turning of the baser metals into gold, and therefore they did not pay much attention to this idea, although this fact was placed on record.

Some time after this we learn that a German named Schultze made copies of drawings with a mixture of chalk and silver nitrate spread on a level surface. The time of this is doubtful, but it was probably about the year 1700. He passed the light, as he says, through translucent paper (made translucent with oil or wax), and objects placed upon the paper left a white impression on the mixture of chalk and silver nitrate—or, as he called it, “lunar caustic.” This was in about 1700, as I said. About fifty years after this time (and indeed it was a little more, it was seventy years, in 1777) Scheele, the Swedish apothecary’s assistant, took up the examination of this horn silver. It seemed to him well worthy of study; and as the result of his work he obtained the first germs that led to the art of photography. But before Scheele could have prosecuted his researches, and before photography could make any important advances, there were two other discoveries in science—and in optics particularly—that had to be made. The first of these was the decomposition of white light, by Sir Isaac Newton, by which he obtained the prismatic colors; that is to say, the colors that we know as violet, indigo, blue, green, and so on down to the red. That was the first step. The next step was the discovery by Baptiste Porter, an Italian, in Naples, which preceded the discovery of Newton (it was about 1590), that a small opening in a dark chamber produced an inverted image on the wall of the chamber. So that between 1590 and 1666 Baptiste Porter and Sir Isaac Newton paved the way for the researches of Scheele upon the action of light upon this simple substance, as they called it, “luna cornea” or chloride of silver. Now Scheele, therefore, at his time, 1777, knew of the discovery of the prismatic colors, or the decomposition of white light by Sir Isaac Newton, and he made the experiment of submitting this horn silver or silver chloride to the action of light after the light had been passed through a prism and he found the light as we know it to consist of violet, indigo, blue, green, yellow, orange and red. Placing the silver chloride in this band of colors, he discovered the important fact that in the red rays the silver chloride received no change—that there was no change made in it. But, as he got along toward the other end of the spectrum, and got into the green and the blue and the indigo and the violet, he found that the color of the silver chloride changed much more rapidly, and he found that the most active in its effect upon the silver chloride were the blue and violet rays. In

addition to this fact he found that the light discolored the silver chloride. Scheele still further proved that the silver chloride was decomposed by the light, and that chlorine gas, or, as he called it, dephlogisticated marine acid gas, was produced. He became acquainted with this previously from his experiments on the mineral braunstein with muriatic acid. So that when he perceived the odor of the chlorine from the decomposition of the silver chloride, he recognized the gas at once, and he says: "When this silver chloride turns black it gives out chlorine," and that was a very important fact. At the red end of the spectrum he found there was little or no effect upon the silver chloride. This was the principle of the camera obscura, and the principle of the camera obscura is the principle of the photographic camera to-day. Practically the photographic camera consists of a dark box, with a hole at one end and at this end there is a place to receive an image. Instead of having a lens there in the front of the camera, as was formerly the practice, it is perfectly possible to get the picture with a small opening, say an eighth or sixteenth of an inch in diameter, and, furthermore, that is the most perfect picture you can get in a camera—a picture without a lens. Now, that is a strange statement, and perhaps in these days it may appear a little wild; but (exhibiting a photo about 5×7) there is a picture made with an opening not larger than a pinhole, and it is a good deal better than many of the pictures taken by the amateurs to-day. This opening being so small necessitates a good deal of time in the action of the light upon the sensitive silver salts behind, and that is the object of placing the lens there. By placing the lens here, instead of having a small opening, you make a larger opening which collects the light in the same manner, brings it to the focus and then the rays diverge again and you get the picture. Now, the rays as they pass through the opening without a lens, begin to diverge as soon as they are in the camera, but with a lens there they are brought together first and then cross and then you get the picture. That is the first step, then, in photography, the production of images by the camera obscura—and that is all the photographic camera consists of—a modification of it. Now, when the facts ascertained by Scheele, *i.e.*, the action of light upon silver chloride—turning it black and producing gas (and by the way Scheele never found out what this gas was and to-day it is a matter of controversy and a problem among chemists)—with the facts ascertained by Scheele, in regard to the action of light Thomas Wedgwood and Vueder made pictures, in 1802. These pictures were very peculiar. They spread upon paper and upon glass

plates that had some gummy material upon them silver chloride—as a precipitate, and then they set their subjects up, so as to get a profile shadow with a strong light upon the surface. Now, where the light passed, of course they got a black mark upon the silver chloride, but the silhouette of the face was in white. Now, that was very remarkable, because they got some very remarkable pictures of which drawings were made. They were white silhouettes on a black background, but remember that the pictures that were thus made, the white silhouettes (if I may use the term) were made by the action of some light. If you wanted to copy them you had to copy them out of the light; otherwise the whole mass would get black, and that was the difficulty. In other words, the white impression could only be examined by candle or some other weak light, and they ultimately became shrouded in darkness and were lost—so we have now none of those pictures.

While they were experimenting in England, a man named Niepse, a Frenchman, was at work upon the same subject—the action of light upon various materials, but in a somewhat different direction. In 1813, or probably before that time, he discovered that certain kinds of bitumen were soluble in oil of lavender, and that when you exposed these pieces of bitumen to some light the oil of lavender would not dissolve them any more. He conceived the idea (how, is not on record), but he thought that if he could coat plates with this bitumen and then expose them to light in a camera he could get a picture upon this bitumen, and where the light had acted the bitumen would be insoluble in oil of lavender. Where the light had not acted that he could dissolve it out. He proceeded to do this, and succeeded in getting pictures upon metal plates. He then, afterwards, etched the plates and thus got a perfect drawing or picture. So he used it simply as a means to produce a picture by etching. Now, understand, using the camera, he obtained an impression upon metal plates coated with bitumen. After exposing the plates in the camera he washed them in oil of lavender and then an etching fluid, and cut the impression into the matter and then they were printed. Some of these pictures are still in existence, they say. I have never seen any of them. After a time the plates were cleaned, and by the help of an etcher's tools or an engraver's tools they were cut still deeper and made very good engraving plates; so that his object was not simply to etch them but to produce plates for engraving.

While this was going on Herschel made an important discovery in 1819, and that was that chloride and bromide and iodide of silver were not soluble when blackened by light. He found that after you had exposed these materials to the light—this silver iodide, bromide or chloride—and had washed all these with hypophosphite of sodium, they would not dissolve. That was important. That made it possible to preserve the silhouette pictures devised or discovered by Wedgewood and Vueder. Therefore, after exposing the plates in the camera, as did Niepse, the Frenchman, he washed them in a solution of hypophosphite of sodium. That took off the chloride of silver that was not acted upon by the light and he preserved the pictures. Some of the first pictures that he made were rather curious. I have not one of his original pictures; I wish I had, but I have a picture made in the same manner. He took a piece of paper and saturated it with salt (he said that he used Bristol drying paper, which was a peculiar paper, made at that time in England). This was soaked in chloride of sodium or common salt, and then it was dipped and had flowed over it nitrate of silver. Therefore he had in the pores of the paper chloride of silver in very intimate contact with the paper. Then he took such objects as ferns and pieces of paper, cut it in various shapes, and laid it on the paper. That produced such an effect as where the objects had laid they had the white impression. If you took this out in the sunlight it would all get black. But he made this important discovery and thus preserved the picture. This was the first photograph made. We do that to-day, and produce other pictures with various other compounds, but I will speak of that later.

In the year 1824 we hear of another Frenchman (now, remember this was a long while ago, in 1819, and we had no photographs yet, although you might call that a photograph (exhibiting the fern picture) yet it is not). In the year 1824 we hear of another Frenchman who was a scene-painter at a theatre in Paris, and he had been using the camera obscura to obtain pictures from nature from which to paint his scenery. That is to say he had a tent built something like that (drawing figure on blackboard) with a lens something like that that was part of a right angled prism, and this light coming from the view, the image was formed in here and spread out upon a table from which he could make a drawing. He used that and was much annoyed at the time it took to get those pictures. He was very impatient, like a great many other Frenchmen. He conceived the idea of “fixing these pictures” as he called them. He did not want to have the trouble of drawing

them. He said: "If I could only find some way of getting that fixed on the surface without the trouble of drawing it it would be a great convenience." This Frenchman was Louis Daguerre, really the father of photography. Now he worked independently for some time, when he met Niepse, the Niepse that had been working on bitumen and oil of lavender, and they formed a kind of partnership in 1829. Now, remember, 1819 was the time that Sir John Herschel had discovered hypophosphite of sodium and its action on these silver compounds. They formed a partnership in order to work out "scene pictures" as they called them. In the year 1833 Niepse died—got tired of the work pretty much, I suppose—and Daguerre continued the work. What his early experiments were we have very doubtful records of. Daguerre did not seem to keep very good records. In the year 1839, little more than fifty years ago, he communicated to the French government a method for making pictures in the camera upon metallic plates. In other words he divulged the secret of the first photographic picture we have—the daguerreotype. This was such a great success and such a wonderful discovery that the French government pensioned Daguerre for his life time, and by an agreement with them the process became public property on August 10th, 1839. Now I have the good fortune to have here to-night the daguerreotype apparatus. This is practically all the paraphernalia of the daguerreotype. First of all was the camera (and you must pardon the condition of it as it is almost forty years old). I know of no other complete set in the United States, so this is rather a relic, and it requires a good deal of care in handling it for it almost falls to pieces (showing the apparatus). Here is where the lens was put and in here is where the plate holder was put. They first had to fix the lenses in the ordinary way with ground glass. Then they had a plate-holder something like ours, that they put the metallic plate in. Now having fixed it, the next thing to do was to present to the sitter this metallic plate, and I have here one of just such plates. Now, into this plate-holder are fitted "kits" as we call them to hold different sized plates. Unfortunately part of this apparatus is lost; *i.e.*, to say all these little details of kits, but they could all be made out of little pieces of wood. Now, the daguerreotype is this: They take a silver-copper plate (a piece of copper plated with silver. When they first did this, they used to solder upon copper plates a piece of silver, then put it in a press and roll it out. After that time, in latter years when the galvanic battery had been discovered and was in common use, they electroplated it). Now, this particular plate was put into a

holder that was held like that. Now the small boy was given one of the buffers or he was put at a wheel that had upon it a backing of felt and on the front of it was chamois leather (it is now long gone on this one—been rubbed off). This plate was then rubbed with a great deal of dexterity and you had to be very careful that you did not scratch it. That was the most important thing about them. It spoilt the picture if you scratched them. They had to be perfectly smooth. As I said, this was sometimes done by holding the plate on a wheel, but the ordinary way was by using one of these buffers. The silver plate was taken out by undoing this screw at the corner. Now, the first thing to do with it, then, is to make it sensitive. It is merely a silver surface now. It was made sensitive by placing it in one of these boxes (showing it) called coating boxes. Now that plate was put into that box (showing the same box), and see there is the lime in the box and it is now probably forty years old, having never been disturbed. In that lime was placed bromine, and it was then covered with a glass cover that fits over this glass trough or dish—it is rather deep. This was then placed with a little pressure—in order to keep the box tight and not let the bromine fumes get all over the studio—and they put the plate in here and pulled this over, so, leaving it there a certain number of minutes, and by action of the bromine vapor it becomes coated with bromide of silver. Then they either put some iodine into this same box or they had an iodine box. After the plate was in there a few minutes, they took it out and put it in there and gave it a dose of bromine. It was found, and by whom I am not sure, that the addition of a little iodine or a small proportion of iodide of silver with iodine of silver gave better effects. So it was then taken out and it was sensitive to light. Now, Daguerre discovered all that. This was then put in the plate holder and exposed in the camera and he got a picture. And it bothered him a great deal, for it faded. If he put that hypophosphite of sodium on it that our friend Herschel discovered, it cleaned the whole picture off. There was not enough of it. So he watched and watched and was weary with making these pictures and having them fade, until he went one day to a closet where he had a lot of these pictures stored, and he was delighted to see that the picture of a certain monument (I think it was) that he had made he thought on that plate some time before, and it was a good picture and a permanent picture. How it came about puzzled him a great deal. In looking around the closet where these pictures were exposed—where these plates were stored—he found that for some reason or other the bottle of mercury had been

broken, and he tried almost every imaginable material in the closet, and at last it struck him it might be mercury. Well, he put some mercury on the plate and he ruined it. "Well, no," he says, "it is not mercury but mercury in a very fine state. I wonder if it is the *Vapor of Mercury*?" He tried it and found that it was. That led to the development of the daguerreotype. Then all he did with a plate was to put it into a vessel with a few drops of mercury, and underneath a little spirit lamp. Then he would put the plate in and watch the heat (some now have a thermometer) and he would just pick it up every once in a while to see how it is developing. That process gave to him the first picture, the daguerreotype, and those are to-day the handsomest pictures ever made by photography. I have two or three of them which are partly spoiled, but to-day they far surpass anything we have ever since done in the science of photography. After the mercury process, it was very easy to wash the plate off. The object of the development was this: that where the light had acted there the mercury seemed to take hold and bring out the picture. Where the light had not acted you could dissolve the silver surface off with cyanide of potassium, which was generally used. But, if you will look at this old-fashioned daguerreotype, you will see that you had to look at them in a certain light; otherwise, you could see nothing.

Sometime afterwards a man named Fitsherbert, a Frenchman, conceived the idea of changing this peculiar picture in silver plate into a gold picture. In other words, he put into the plate a little chloride of gold and produced a daguerreotype which can be seen pretty clearly by looking squarely at it.

The beginning of the daguerreotype flourished only a short time. While Daguerre and others were working at the daguerreotype, Fox Talbot, a rich Englishman, took up the subject from another point of view. He conceived the idea of making a negative. Of course, every picture you took by Daguerre's method you had to make a sitting for it. Such are the pictures up in the School of Mines of William Lloyd Garrison and Daniel Webster. They had to sit right down in front of the box, and copies could not be had. That was the trouble with the daguerreotype. You had one picture for every sitting. To make the difference between the positive and negative more clear, I have brought here to show you to-night (producing them) some positives and negatives printed on the same piece of paper. When the picture comes out of the camera and the plate is developing (exhibiting it) that is what it looks like—where the light struck all the light parts of the

picture are black, and where the light did not strike all the black parts of the picture are white. If I take the same surface, containing the bromide of silver, iodide of silver or chloride of silver, and place it underneath that and expose it to the sunlight, where the light strikes through it will produce black, just as in the original object, and when I get through I get the positive. So there is a negative and there is a positive from the same picture. Now, that was Fox Talbot's idea. He says "If I can do that, I can make pictures *ad libitum*." With this object in view he coated paper with silver chloride. He exposed it then in the camera, fixed it in a solution of salt—common salt or iodide of potassium—and when he got through the picture was a permanent one, because the iodide of potassium dissolved out the white parts that were not affected by the light. From this negative he obtained other prints.

Now, various modifications of Fox Talbot's process, were brought out, and a man named LaGray, I think (or at least it was just about the time he lived) conceived the idea of making these pictures more transparent by waxing them. That was the first good negative we had. It was a modification of Fox Talbot's idea, only he waxed the paper. Then about the same time it was found that a mixture of chloride of iron and cyanide of potassium, when mixed together were acted upon by light. Herschel discovered this, and that was the way we obtained the blue print, which is far older than the photograph. Sir John Herschel found that a mixture of chloride of iron and cyanide of potassium, when exposed to sunlight made Prussian blue. So that if you take paper and coat it with this mixture and then expose it under a negative you get a blue picture.

The trouble with these paper pictures was that you could not eliminate the grain of the paper, and if you will examine these close enough you will see that they are blurred. This one printed from that particular negative is blurred—very much blurred. These sensitive silver compounds are so sensitive that the grain of the paper produces an impression. Now, in 1848, Niepse, a nephew of the first Niepse, thought it would be a good idea to use glass plates coated with albumen. He took chloride or iodide of silver, mixed it with white of egg, spread it on plates, heated the plates, which, of course, coagulated the albumen, and that fixed his film upon the glass plates. That was quite a step. Now, we had gotten rid of the paper. By the way, I made a little mistake there about the way he got the picture. He got

the picture by putting salt in the albumen and then coagulating it, and then he dipped the plate into a solution of silver nitrate and in that way got the precipitate in the film itself. This was important but troublesome and not always successful.

Now, a few years before another discovery was made. Remember that this was in 1848 that Niepse worked with the albumen process. In 1840, Schurben, a Swiss chemist, discovered gum cotton. This gum cotton is a nitrated compound of cotton, made by the action of concentrated sulphuric and nitric acids upon cotton. Sometime afterwards Maynard, a Yankee, in Boston, discovered that this gum cotton was soluble in alcohol, and ether, and then he found that by evaporating the substance he got the thin film of collodion. Scott Archer, an Englishman, conceived the idea of using this film as a vehicle for these particularly sensitive silver salts for photographing. His method was pretty much that which is followed to-day and that is still in use to quite a large extent.

In this process we have this series of operations: First, the plate must be perfectly clean. That is essential. Any little spot upon it will form a nucleus which will spread over the surface of the plate. The plate is then coated with albumen and allowed to dry without heating. It is then flowed with this collodion, and in the collodion is put the chloride, iodide or bromide of silver, which you need. It is generally the chloride, iodide or bromide of silver. This collodion is afterwards dipped into a silver bath, and then we get the sensitized silver surface, very thin and perfectly transparent. It is then ready to go into the camera. It is put into the camera soaking wet with nitrate of silver. It is exposed and then developed with a solution of sulphate of iron with some acetic acid. After it is developed, the developer is washed off, fixed with hypophosphite of sodium, dried, varnished and we get the negative.

Now, the curious part about this wet plate process is that it is slow. The compounds are not very sensitive compared with the modern compounds. In the second place it is essential to use it wet. If you took the plate out of the silver bath where you sensitized it and washed off the nitrate of silver adhering to it and put the plate in the camera you would not get a picture. The silver nitrate is essential to the production of the picture. It acts in this way: Where the light has acted upon the sensitive silver compounds and you proceed to develop the picture, when you mix the sulphate of iron and

pour the developer upon the plate, as the iron comes in contact with the nitrate of silver, with which the plate is wet, it produces metallic silver, which adheres to those parts of the picture which have been acted upon by the light. That seems to be the philosophy, because if you wash the nitrate off you cannot develop a picture upon such a plate.

Now, this process of photography revolutionized the daguerreotype, revolutionized photography and the daguerreotype became obsolete. I think it displaced the daguerreotype in three years. This process was such an advantage—collodion was such a nice substance to work with—that it revolutionized the photography of those days, and the daguerreotype fell out of existence.

Now, when you take into consideration the time that people had to sit for their pictures—five or six minutes—you can conceive how hard it was to keep still. They had such queer contrivances to keep the head straight, they screwed you up in various positions, and this was particularly exasperating where they had to take pictures requiring a good deal of time. Dr. Draper, who took some of these daguerreotypes, and who I believe was the first photographer of these pictures, desired to take a photo of his estimable lady. His studio was in the old University Building in Washington Square. I believe Mrs. Draper had to sit twenty minutes for that picture. In order to produce the best effect he had a tank made in the top of the laboratory so as to produce a blue light. Mrs. Draper was very patient while he was at work with this, and unfortunately, Dr. Colton tells me, the result was two pictures on the same plate. I should think it would. That was the first effort ever made to take the human face with the daguerreotype. Of course, with all that paraphernalia, with that slowness of action, anything that worked within a minute was considered wonderful, and that was practically what happened when Scott Archer discovered collodion.

This wet plate process continued from 1851 to 1871, about twenty years. I have the pleasure of showing you an amateur outfit for this process, used in 1860 to take to the Rocky Mountains (exhibiting it). That is an amateur outfit carried over the Rocky Mountains in 1860 to take pictures. Here is the old tank that carried the water. Here are some of the bottles of chemicals, and the way it was managed was this: This was hooked up, on the end of these sticks. This was the black cloth used as the developing room by the operator. Here is a little window with yellow glass to develop

the pictures. The plates and bromide of silver was carried in these two boxes. That was carried on top of the mule and the boxes on the sides of the mule, so that he had a pretty good mule.

Now, to-day we do the same work with that apparatus (exhibiting apparently a Kodak), and a great deal better work it is.

In 1871 a more important revolution took place even than the wet plate process or the daguerreotype. Many efforts had been made to overcome the use of the wet plate—the plate wet with nitrate of silver, and some of the efforts were very successful but usually troublesome. The plate was kept moist in a variety of ways: by honey, by tea, by infusion of tea, by beer, by coffee, and a multitude of all the funniest concoctions you could think of, but the process was destined to fail.

In about 1870 it was conceived that you could make an emulsion of these peculiar compounds of silver—these sensitive silver compounds—that you could make an emulsion that you could pour upon the plate and produce a picture just when you pleased, and it was found that by mixing the chloride that produces the sensitive material in one portion of your collodion and putting nitrate of silver into another portion of the collodion, in certain proportions, you could produce a collodial emulsion. They had to be mixed in just exactly the right proportions, so as not to have an excess of nitrate of silver or an excess of bromide.

But that process failed and only lasted a few years; although I have here one of the plate holders used by such a process.

This was between the time of the wet plate process and the modern dry plate, when they used collodial bromide emulsion. It was a kind of a compromise between the wet plate and the dry plate. In 1871, Dr. R. L. Maddox, of Bath, England, had the idea that he would use gelatine, instead of albumen or collodion, as a vehicle to hold these silver salts upon the glass surface, and he found, among other things, something that surprised him—that when he put the silver salts in to contact with this gelatine they became wonderfully more sensitive than ever before.

The idea is this: That you make a gelatine mixture of a certain strength—the proportions required a certain amount of soft gelatine and a certain amount of hard gelatine. Into that gelatine you pour, with constant stirring;

you pour a mixture at the same time—some particular bromide, generally bromide of potassium and nitrate of silver—in a very thin stream and keep it thoroughly stirred up. If you go too fast, you will not get the right result; but the result is, when you get through and do it right, you get a beautiful milky fluid, and that fluid contains bromide of silver in a wonderful state of suspension—very thin—and it remains suspended in this fluid. Now let that set—this cream or “emulsion,” as they call it—and you have as a result iodide of silver and iodide of potassium. You let the emulsion set and it produces a jelly, that jelly is then cut up into shreds, rubbed through a sieve or something of that kind to make it thoroughly divided, and washed thoroughly with water. Having done that it can be melted, and if you melt it and heat it to a certain temperature, there does not seem to be any limit to the sensitiveness of the material. If you use it cold it requires a second or two to produce a picture. If you cook it, however, you will find that it will become more and more sensitive to light, until it is actually possible to take a picture of a projectile traveling four hundred metres per second. I have such a picture. The only trouble is that some of the plates made are so sensitive to light that we cannot get a light non-active enough to develop them. Having these bromide plates then in the camera—this sensitive material coated on these glass plates in the camera—you have got to be very careful that the light does not get to them. The consequence is that the plate holders are made with extreme care.

The result of this gelatine-bromide of silver process is this: that we can have plates in packages. We can put these emulsion plates and carry them off where we please, and, what is still more important, we can put the emulsion upon very thin material, and I have here (exhibiting them) thin sheets of celluloid upon which this emulsion has been spread and pictures taken. That is not all, either; they can make it still thinner (producing small camera) they can put it on a roll and in this camera is one of those rolls, and in that box I can take a hundred pictures without reloading the instrument. The way it is done, I, when I want to produce a new surface, simply wind the old one off with this winding machine. There is an opening at the front of the camera. Press just below this, so, and you have the picture. Now just wind the film off and you are ready for the next picture. Now pull it again, and this is so easy that some manufacturers say: “You simply push the button and we do the rest for you.” That is nonsense, they don’t do the “rest” for you. A friend of mine took one of these to Europe, and with it a

dozen rolls of film, all of which he used. When he returned he sent them to the manufacturers and I think he got about twelve pictures back. Not every time you press the button is a good picture produced. You have to know a little bit about the science and use a little judgment.

Such is the state of photography to-day that this material can be spread upon any kind of transparent surface. In the case of plate, they are put in holders like this, generally only two on each side, and slipped into this frame in a dark room, in which no light can be used except one emitted through a deep red chimney. (The professor here exhibited such a chimney.) Then, the material that is used for developing these pictures is somewhat different from the old method. We use organic compounds, alkaline solutions, and organic matters capable of taking up oxygen. These organic materials, in conjunction with some alkali, are capable of taking up oxygen. They produce a disoxygenizing action. After dipping, that gives you the negative.

The prints are made in a variety of ways. The facility with which these apparati can be used has led to an enormous variety. You can have an apparatus something like that, or something like this, which is smaller.

In the United States there are to-day probably about ten thousand professional photographers and thirty or forty thousand amateurs, who usually do nothing but spoil plates. To give you an idea of some of the work done, not altogether by professionals, I have picked out from the number of pictures I have a few samples of the work. Here is a picture of a cattle ranch in Colorado. I have one a little larger of a horse race, but this is about as large as they can be made. That will give you an idea of the instantaneous effect. The distance between the foot and the top of the mountains is about twelve miles, so that you can get an idea of the capacity of the camera, of the sensitiveness of these compounds. Here is a Mexican picture which shows the great beauties of the Mexican flora—the cacti. Here is a study “King Lear” made by Buffler, the photographer. That is about as large as you can get. It is a pretty large plate to handle. Then there is another study “The Five O’clock Tea” some ladies at tea, by the same man as “King Lear.” Here is another study, “A Game of Sixty six.” Those are all silver prints, made with chloride of silver, using glass negatives and producing the positives by having the chloride of silver in albumen. The best vehicle to-day for making positive prints is albumen with chloride of silver.

It is found that if you take a mixture of gelatine and bichromate of potassium, and put into the mixture some pigment and expose it under a negative where the light acts, the gelatine is made insoluble and holds the pigment, and where the light does not act the gelatine is still soluble and can be washed away. Here is such a picture and it is very interesting—"In Camp." The shadows in that picture are on the white paper underneath.

Here are a couple of pictures of silver, two Bavarian pictures. This one, of a little girl, is by Einlander of Cologne, instantaneously taken without a head-rest, which is a very difficult piece of work. This is the same idea, instantaneously taken. Here are two pictures very interesting, which were in the exhibition at Chicago. They are pictures in platinum, showing that we are not confined to simply silver salts. We have here in this last picture one of the chlorides of platinum, the platine chloride. It cannot be spoiled in any way. The picture is good as long as the paper is good.

Here is an example of a yacht picture. It is the English yacht Iris. It is a fine picture. The yacht is travelling very fast.

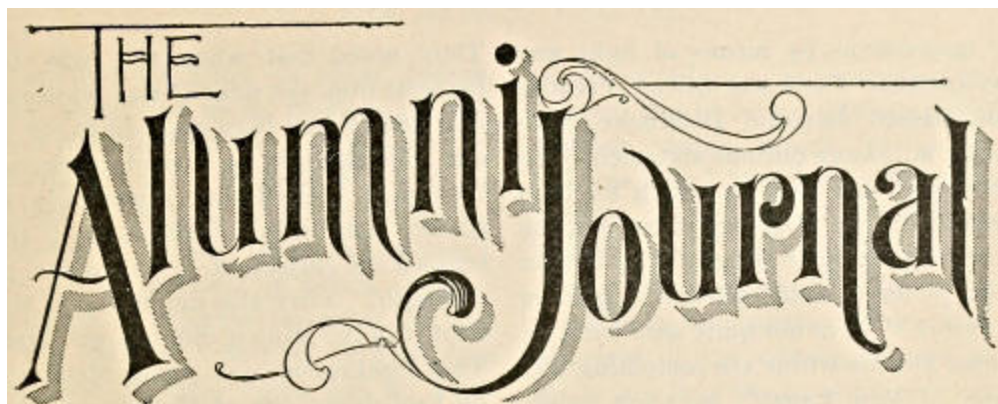
Here is a picture on the East River, made by Dr. Habershaw, showing the work of amateurs in this line.

I could tell you a good deal more about this subject, but there is only one other thing I would now like to mention. Some of you, I suppose, have heard a great deal about taking photographs in colors. We are very near it. They have produced in France, Germany and England pictures of the spectrum in the silver salts: that is to say, with the colors of the spectrum. They are very weak and have to be looked at in a certain light. They are the result of interference of the thin films. We are doing something more important. We are learning to make the whole spectrum. For example, we can to-day get just as good an impression upon silver salts with a red light as Scheele did with a violet light in 1774. That leads to what is called orthochromatic photography, that is photography that will give us every color in the spectrum. It has been found possible to make pictures in certain colors. A long time ago, the spectrum was separated into three colors, red, yellow and blue of certain kinds.

Now, if you take a picture in a red light of a certain character, and another of the same subject in a yellow light of a certain character, and another in a blue light of a certain character, you have three negatives. You can make

three negatives, one of the red light, one of the yellow light and one of the blue light. Now, by taking pigments and printing in a press like a lithographic press, you can make a red positive from the red negative, and a blue positive from the blue negative and a yellow positive from the yellow negative, and in that way you may get three impressions, which is the result in the same colors. You must not stop there, however. There is a certain amount of shadow, and the result of it is that they have to what they call “over-lay,” taking the three colors separately and superimposing them in printing. Remember, the red parts of the picture are taken with the red light. That is, suppose you put a red piece of glass in front of your camera, then only the red parts of the picture pass through to the sensitive plate. Then repeat the operation with the blue glass and the yellow glass, and the result will be as above.

Now I hope I have not bored you by any profuse details. I did not intend to. I only tried to interest you in one of the most important inventions of the Nineteenth Century. The steam engine, the telegraph, the telephone and the photograph are four of the grand inventions which the century has produced, and I think every intelligent person should learn something about them. I am afraid that I have had too little time to do the subject justice. You can understand how much more there is behind this superficial view. I only have to thank you for your very kind attention.



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THE ABILITY OF CONSTRUCTION.

At this stage of the world's history men of ability and even of genius in a certain sense are not rare. The result is that in all of our institutions of learning the requirements become more stringent and by the time graduation arrives we see the survival of only the very best men. We find the same classes of men throughout life that we find in college—we find men of energy and slothfulness, men devoted to pleasures and by nature politicians, men of ability of construction and men of power in criticism. While at College the training to-day is chiefly analytical and the result is that men are prone to examine everything closely and some even learn to take delight in tearing things to pieces. There are some men who are utterly ruined so far as their inward happiness and that of those about them is concerned by their critical tendencies. They do this to the detriment of their own energies and abilities of construction and hence never or but seldom build anything, but employ their days in tearing down what others have built. The critic is necessary and essential in every department of labor where human thought is allowed entrance. Criticisms that are honest always help the builder and are a gain to posterity.

It is questionable if it is desirable for the conscientious young man to encourage in his life a too critical tendency. It is not necessary to look at the bright side of the affairs of life, or even to look upon men charitably, so to speak. It is sufficient for every young man especially to look upon events of life as they are. It is decidedly important for the man of aspiration to look upon life with its duties when he has had sufficient rest and food and exercise. Wrongs may be righted and errors corrected in but two ways: the thoughtful way and the thoughtless way. The thoughtful way is always attendant of health and with a broad minded and large hearted individual. It is not our desire, however, to dwell too long upon the subject in the abstract as we are anxious to reprint the closing words of Senator Henry Cabot Lodge's Phi Beta Kappa oration delivered last June at Harvard College. He said in closing:

“How then is a university to reach the results we ought to have from its teachings in this country and this period? Some persons may reply that it can be obtained by making the university training more practical. Much has been said on the point first and last, but the theory, which is vague at best, seems to me to have no bearing here. It is not a practical education which we seek in this regard, but a liberal education. Our search now and here is not for an education which shall enable a man to earn his living with the least possible delay; but for a training which shall develop character and mind along certain lines.

“To all her students alike it is Harvard’s duty to give that which will send them out from her gates able to understand and to sympathize with the life of the time. This cannot be done by rules or systems or text-books. It can come from the subtle, impalpable, and yet powerful influences which the spirit and atmosphere of the great university can exert upon those within its care. It is not easy to define or classify these influences although we all know their general effect. Nevertheless, it is, I think, possible to get at something sufficiently definite to indicate what is lacking and where the peril lies. It all turns on the spirit which inspires the entire collegiate body, on the mental attitude of the university as a whole. This brings us at once to the danger which I think confronts all our large universities to-day, and which I am sure confronts that university which I know and love best. We are given over too much to the critical spirit and we are educating men to become critics of other men instead of doers of deeds themselves.

“This is all wrong. Criticism is healthful, necessary, and desirable, but it is always abundant and infinitely less important than performance. There is not the slightest risk that the supply of critics will run out, for there are always enough middle-aged failures to keep the ranks full if every other resource should fail. Faith and hope, and belief, enthusiasm, and courage are the qualities to be trained and developed in young men by a liberal education. *Youth is the time for action, not criticism.* A liberal education should encourage the spirit of action, not deaden it. We want the men whom we send out from our universities to count in the battle of life and in the history of their time, and to count more and not less because of their liberal education. They will not count at all, be well assured, if they come out trained only to look coldly and critically on all that is being done in the world and on all who are doing it. We cannot afford to have that type, and it

is the true product of that critical spirit which says to its scholars: "See how badly the world is governed; see how covered with dust and sweat the men who are trying to do the world's business, and how many mistakes they make; let us sit here in the shade with Amaryllis and add up the errors of these bruised grimy fellows and point out what they ought to do, while we make no mistakes ourselves by sticking to the safe rule of attempting nothing." This is a very comfortable attitude, but it is one of all others which a university should discourage instead of inculcating. Moreover, with such an attitude of mind towards the world of thought and action is always allied a cultivated indifference than which there is nothing more enervating.

"The time in which we live is full of questions of the deepest moment. There has been during the century just ending the greatest material development ever seen. The condition of the average man has been raised higher than before, and wealth has been piled up beyond the wildest fancy of romance. We have built up a vast social and industrial system, and have carried civilization to the highest point it has ever touched. That system and that civilization are on trial. Grave doubts and perils beset them. Everywhere to-day there is an ominous spirit of unrest. Everywhere is a feeling that all is not well, when health abounds, and none the less dire poverty ranges by its side, when the land is not fully populated and yet the number of unemployed reaches to the millions. I believe we can deal with these doubts and rents successfully, if we will but set ourselves to the great task as we have to the trials and dangers of the past. But the solution will tax to the utmost all the wisdom and courage and learning that the country can provide. What are our universities, with their liberal education to play in the history that is now making and is still to be written? They are the crown and glory of our civilization, but they can readily be set aside if they fall out of sympathy with the vast movements about them. I do not say whether they should seek to resist or to sustain or to guide and control these movements. But if they would not dry up and wither they must at least understand them.

"A great university must be in touch with the world about it, with its hopes, its passions, its troubles, and its strivings. If it is not it must be content.

‘For aye to be in the shady cloister mewed,
Chanting faint hymns to the cold, fruitless moon.’

“The university which pretends to give a liberal education must understand the movements about it, see whether the great forces are tending, and justify its existence by breeding men who by its teachings are more able to render the service which humanity is ever seeking.”

Professor Fried. Aug. Flückiger died on Dec. 11, 1894, at Berne. He was the foremost pharmacognosist and scientific pharmacist of his time. An extended account of his life and works will appear in a later issue of THE ALUMNI JOURNAL.

NEW LITERATURE.

Readers desiring any of the works contained in this list can obtain them through B. Westerman & Co., 812 Broadway, Gustav E. Stechert, 810 Broadway, or other foreign booksellers.

Bacteriology.

Mikrophotographischer Atlas der Bakterienkunde.—C. Fraenkel u. R. Pfeiffer. 2 Aufl. 11, u. 12. Lfg. Berlin: August Hirschwald.

Mikrophotographischer Atlas der Bakterienkunde.—Itzgerott u. Niemann, Leipzig: J. A. Barth.

Botany.

Beitrag zur Kenntniss der Bestandtheile von Cnicus benedictus mit hauptsächlichliche Berücksichtigung des darin enthaltenen bitter schmeckenden Körpers.—Karl Schwander. Inaug.—Diss. Univ. Erlangen.

An examination of the constituents and particularly the better principle of *Cnicus benedictus*.

Beitrag zur Kenntniss des Bitterstoffes von Citrullus colocynthis.—Rud. Speidel. Inaug.—Dissert. Univ. Erlangen.

Weitere Beiträge zur Chemischen Kenntniss einiger Bestandtheile aus Secale cornutum.—Hans Zeeh. Inaug.—Diss. Univ. Erlangen.

Uebersicht der Leistungen auf dem Gebiete der Botanik in Russland während des Jahres, 1892.—Zusammengestellt von A. Famintzin u. S. Korshinsky unter Mitwirkung von Anderer. Aus dem Russ. übers. von F. Th. Köppen. Leipzig: Voss. A review of the history and events in botanical works in Russia during 1892.

Atlas der officinellen Pflanzen.—A. Meyer u. K. Schumann. 1892-1894. Leipzig: A. Felix. Darstellung und Beschreibung der in Arzneibuche für das

Deutsche Reich erwähnten Gewächse. Zweite verbesserte Auflage von "Darstellung und Beschreibung sämtlicher in der Pharmacopœia Borussica aufgeführten officinellen Gewächse von O. C. Berg u. C. F. Schmidt."

Chemistry.

A Text-Book of Organic Chemistry.—A. Bernthsen. Translated by G. M'Gowan. 2d Eng. Ed. Revised and Extended by the Author and Translator, London: Blackie.

Chimie médicale.—Corps minéraux. Corps organiques. L. Garnier. Paris: Rueff et ciè.

Nozioni di Fisica. Chimica e Mineralogia ad Uso delle Scuole tecniche e delle Preparatorie alle Normal.—M. Borzone. Torino.

Grundzüge der mathematischen Chemie.—Georg Helm. Leipzig: Wm. Engelmann. The author discusses the transformation of energy by reason of chemical action.

Kurzes Repetitorium der Chemie.—1. Theil Anorganische Chemie. 2. Aufl. Ernst Bryk. Wien: M. Breitenstein.

Grundzüge der Chemie und Mineralogie für den Unterricht an Mittelschulen.—M. Zaengerle. 3. Aufl. Munchen: J. Lindauer.

Hygiene.

Text Book of Hygiene.—G. H. Rohe. Philadelphia: F. A. Davis Co.

A comprehensive treatise on the principles and practice of preventive medicine from an American standpoint.

Materia Medica.

Organic Materia Medica and Pharmacognosy. Illustrated. By Prof. L. E. Sayre: P. Blakiston & Co., Philadelphia.

In these days of degenerate rivalry among educational institutions, and particularly among the different classes of technical schools, when their

officers are wont to prefer the very poorest of text-books, written by one of their own number, for the best of them should it emanate from a rival institution, we have become accustomed to looking upon publications of this sort as serving merely, like an electoral vote, to count one among the general collection. It can scarcely be expected that text-books written from such standpoints and with such motives can have much permanent value, and the future educational historian will doubtless look with amazement upon the trash of this character which has been brought to light during the present era. In the midst of this wearisome train of events it is refreshing to have presented to us a new text-book, whose publication constitutes, as to its main part, a real event in the history of pharmaceutical education.

Prof. Sayre's work on Pharmacognosy has a real reason for existence in its scope, arrangement and execution. It is new and original, and will stand by itself as a prominent American text-book. If it possesses glaring and in some respects fatal defects, it at the same time presents the merit of ingenuity in construction as well as in the selection of subject matter, and it cannot fail to become a much-used reference book, not only by the pharmaceutical profession for whom it is intended but by physicians as well. It is perhaps unfortunate that so many individuals, and nearly all of them students, should have been given a free hand in the working out of the various departments, and that their products have not been in all cases perfectly harmonized by the master. It is also unfortunate that so many statements should have been taken, without investigation, from other authors. A brief scrutiny of the pages will suffice to reveal this composite origin, even if one does not read the acknowledgments of the author in his preface. Doubtless Prof. Sayre, while he has not greatly interfered with the individuality of presentation of these different subjects, has taken pains to verify the accuracy of the facts and conclusions recorded. Should such prove upon closer investigation to be the case, the defect referred to must doubtless be considered as one of style merely.

The appearance of an American work on Pharmacognosy is of so much importance that it is not inappropriate that it be analyzed with some degree of fulness. The book consists of two parts with three appendices. Part 1 is on "Pharmacal Botany," while part 2 is upon "Organic Materia Medica and Pharmacognosy." It is impossible to review this work fairly in the interest of the public as well as of the author without recording the opinion that the

eighty-two pages comprising Part 1 should never have been published, if we regard either the reputation of the author or the welfare of students of pharmacy.

Our American text-books on Pharmaceutical Botany, (not “Pharmacal Botany,” as the author unhappily calls it, which would mean the Botany of the Pharmacy, or of the place in which pharmaceuticals are practiced,) bears no evidence that any author has yet comprehended the needs of pharmaceutical students in this direction, or has adjusted his instruction so as to accomplish the object for which it was devised. The idea invariably indicated by the writings, even if not intended by the writers, is that as the application of botanical knowledge to the practice of the pharmacy is limited, its teachings may therefore be superficial, indefinite and vague. The true idea it seems to us is, that it should be curtailed and limited only as to the portions of the field covered; but these requisite portions should be taught with a fulness of illustration, a clearness of presentation and a simplicity of style, all the more marked because the student is deprived of the enlightening effect contributed in other cases by those portions which are here necessarily omitted.

As a synopsis, or summary of knowledge, intended to guide the teacher instructed in the subject, these eighty-two pages will answer fairly well; but to enable a student who is proceeding *de novo* to gain a knowledge of structural botany for the purposes of pharmacognosy, we can see nothing but failure. Herein we criticise the book, not specifically the author. Publishers’ books are not always authors’ books. It is doubtful if any publisher can be found willing to publish as a business enterprise, a perfect text-book of Botany for pharmaceutical students.

When such appears, it will be as a labor of love, by one whose regard for the subject is such as to lead him to donate his time and labor, and whose means enable him to bear the burden of a financially unsuccessful enterprise.

The part of the work under criticism is a mere series of definitions, illustrated in a highly unsuccessful manner, and frequently losing sight of the requirement that a definition must include the whole of the thing defined and nothing else. It is very naive to say: “All organic matter containing a green coloring matter called chlorophyl, belongs to the

vegetable kingdom,” without directly stating that no other class does, which statement would leave out the fungi, a part of the definition of which is that they contain no such matter. To define Morphology as treating—“Of the organs of plants and their relations to each other,” is not to define it at all, as that would include the whole of Organography, and does not even exclude Physiology, except by virtue of the author’s preceding clause. Systematic botany, defined as “That division which treats of the arrangement and classification of plants,” does not suggest the vital characteristics of that subject. It would be more philosophical to refer to the distinctive characteristics of Phanerogams as the manner in which the embryo is produced within a true seed, than to intimate that the embryo is entirely foreign to cryptogamic reproduction. These definitions, taken from less than two pages of matter, indicate to our mind a lack of the expenditure of time requisite to bring forth a set of new definitions more perfectly in accord with the fullest knowledge of to-day than any list which has yet appeared; and yet when the instruction given in a new text-book is chiefly limited to definitions, that is the very least that should have been attempted.

Some of the morphological definitions are actually at variance with accurate descriptive usage, as that of primary and secondary roots, duration, etc. To call a stem an “axis” and a root an “axis” of a different kind, is to perpetuate a term at the expense of all regard for that accuracy which is the most important element of scientific language. Such subjects as venation are of prime importance to the pharmacist, and so far from restricting the teachings to several of the more important terms presented in ordinary text-books on botany, the classification should be elaborated in its fullest details. Compare the definition of classes, as “Plants resembling one another in some grand leading feature,” and of orders or families, as “Plants that very closely resemble each other in some leading particular,” with the clear presentation of ranks in class characteristics, given by Agassiz a generation ago, and which should, if anything, have been improved upon in the light of modern knowledge and perfected usage.

The subject of nomenclature, the recent agitation of which has done more to expose and shatter erroneous practices in scientific thought and custom than any other influence, and whose correct apprehension is the very corner-stone of pharmacopœial definition, we do not see anywhere treated.

It is a pleasure to turn from a contemplation so depressing to the spirits of one who has labored hopefully for years to secure a just and rational treatment of his favorite study at the hands of Pharmaceutical educators, to Part II. of Prof. Sayre's book, a work so bright and practical, so replete with new and helpful ideas in the teaching of practical Pharmacognosy, and so full of information, both standard and exceptional, though unhappily marred by many errors, as to secure for it at once a prominent place upon the shelves of the "Handy Book Case."

The principle is here adhered to of making a single volume do duty as a text-book of Pharmacognosy and of "Materia Medica," as the latter term is commonly used. We have never looked upon this method as being practicable, but Prof. Sayre resorts to a most ingenious device never before resorted to, by which it must be admitted that better results have been obtained than have previously been reached. What might be called a "Pharmacognostical Key," or a synopsis of Pharmacognosy, is presented separately in advance of the main body of Part II. Here the drugs are numbered to correspond with the consecutive numbering prominently displayed under the second arrangement, that by natural orders, the proper method for retaining and displaying the natural relationships of active constituents and medicinal properties. The "Pharmacognostical Key" appears to us a failure in its practical workings, owing to indefinite characterization, by reference to taste only of the headings. If a drug is both bitter and aromatic, we have to look for it both in Class I. and Class III. A bifurcating key is here required, or better, we might take a combination of characters for each heading. On the whole, this key, while elaborate and very full, and subject to great improvement by a few trifling changes, we must regard as inferior to that of Maisch's text-book. Prof. Sayre very sensibly omits all attempt to classify volatile oils, except by indicating their sources.

The arrangement of the matter of the second part is, first, a brief description of the ordinal characters, followed by a list of the drugs belonging to that order, those official in heavy-faced capitals; then the drugs are taken up separately, the official names and synonyms in the important languages presented, the definition, botanical characteristics, sources, related, and similar articles, description of drugs, with the more important characters printed in heavier type, accompanied generally by a picture of

the plant and of the drug, gross and structural, important constituents, actions and uses, and a synopsis of the official preparations. The doses of the drugs are given, but not of the preparations, though the strengths of the latter are stated. An unfortunate feature, as in Part I., is the illustrations. They are not at all uniform in effect. While the method followed has given exceptionally good results in some cases, yet in many others they are very unsatisfactory, and this is more particularly true from a scientific than from an artistic point of view.

Valuable a contribution as is Part II., there is an evident unfamiliarity with, or disregard of, the commercial aspects of drugs. For instance, the important distinctions between *Cassia vera* and *C. lignea*, and the subject of Batavian Cassia, a correct understanding of which is a great aid in the economy of the drug store, are entirely omitted. The distinctions between Coto and Paracoto are not clear, and in the facts concerning commercial occurrence are reversed. Mace is not, as described, a “membrane,” neither does it “invest the kernel.” Moreover, nothing is said about Wild Mace, now so extensively used as an adulterant that it is possible that it constitutes the larger part of commercial Mace. “Reddish brown” boldo leaves are old and worthless. The description of *Piper longum* is only partly true, according to the variety under consideration, and the individual parts are not “berries.” The part rubbed off from *Piper album* is not correctly described as an “epidermis.” The important characteristics distinguishing true from false cubebs is not given.

Appendix “A” is a valuable contribution on the subject of insects injurious to drugs.

Appendix “B” is no less important, it being an account of the contributions of organic chemistry to *materia medica*.

Appendix “C” treats of “Pharmaceutical Microscopy” in such a fragmentary and superficial way that it will scarcely be found of service to any one in these days.

H. H. RUSBY.

Pharmacy.

Einführung in die Maassanalyse.—M. Vogtherr. Für junge Pharmaceuten zum Unterricht und zum Selbststudium. Unter Berücksichtigung des Arzneibuches für das deutsche Reich und der Ergänzung desselben durch die ständige Commission für die Bearbeitung dieses Arzneibuches. 2. Aufl. Newied: Heuser's Verlag.

Pharmaceutisk Haandboog for 1895.—E. P. F. Peterson. Kjobenhaven: F. Host & Sons.

Photo-Micrography.

See also Bacteriology.

Photo-Micrography.—H. van Heurick. Eng. Ed. Re-edited and augmented by the author from the 4th French edition and translated by Wynne E. Baxter. With Illus. London: Crosby, Lockwood & Son.

Photography.

Deutsches Photographen Kalender.—K. Schwier. Taschenbuch und Almanach für 1895. 14. Jahr Weimar.

Physics.

Manual of Physico-Chemical Measurements.—W. Ostwald. Translated by James Walker. London and New York: Macmillan.

A Laboratory Manual of Physics and Applied Electricity.—E. L. Nichols. 2 vols. London and New York: Macmillan.

Anfangsgründe der Physik mit Einschluss der Chemie und Mathematischen Geographie.—K. Koppe. 20. Aufl. Ausgabe B in 2 Lehrgängen. Für höhere Lehranstalten nach den preuss. Lehrplänen von 1892. Bearbeitet von A. Husmann. II. Th.: Hauptlehrgang. Essen: G. D. Baedeker.

Elementi di Fisica ad Uso delle Scuole secondarie.—F. Cintolesi. Livorno.

Thermo Dynamics treated with Elementary Mathematics.—J. Parlseo.
London: S. Low & Co.

THE MOST RECENT WORK.

A Seidlitz Powder.—A. Gunn made an examination of some powders and found the blue powder to consist of magnesium sulphate and sodium bicarbonate. The white powder consisted of tartaric acid. Evidently there had been a mistake or else it was a bold attempt to cope with the cutting system and its cheap prices. One wonders that the makers should expect the unusual effect of trying to dissolve the contents of the blue paper to pass unnoticed.—*Pharm. Jour. Trans.*, 1894, 534.

Ointment of Mercuric Nitrate.—C. H. La Wall (*Amer. Jour. Pharm.*, 1894, 525). The following fats have been suggested as a substitute for the lard oil: Neatsfoot oil, lard, butter, peanut oil, almond oil, castor oil, palm oil, bear's oil, ox marrow, beef suet, stearic acid, petrolatum, and almost all of the other fats from the animal and the vegetable kingdoms, and even one from the mineral kingdom, appear to have been experimented with in the vain hope of finding some fat or oil which would make a good and durable ointment.

Several writers have taken another course and have tried to preserve the products obtained from former processes. One advises keeping the ointment in a jar and covering it with a layer of glycerin to prevent oxidation; others have tried the addition of camphor; still others have given their attention to the mercurial portion of the ointment, and suggest making the nitrate from the oxide of mercury instead of making it from the metal. Some have even been skeptical as to the reliability of any process, but those who have approximated the truth more nearly are they who advise careful manipulation, especially as regards temperature.

The author employs the official ingredients and quantities and heats the lard oil to 100° C., removes heat, and adds the nitric acid without stirring and reapplies heat when effervescence ceases until all gas is expelled. It is best to use a vessel of six times the capacity of the quantity to be made to allow for the copious effervescence which takes place. When the foregoing mixture has cooled to 40° C., the solution of mercuric nitrate is added and the temperature is raised gradually to 60° C., and maintained until no

further evolution of gas is noticed. If it is then agitated until cold, as usual, the resulting product will comply with the requirements of the Pharmacopœia.

Ointment made by the U. S. P. method, which has become spongy, may be remedied by elevating the temperature to 60° C. and cooling with agitation.

Typical Bacilli.—E. Klein [*Quart. Jour. Micros. Sci.*, 1894, 1-9 (1 pl)] concludes from observations on the bacilli of anthrax diphtheria, and tubercle, that these species are not such typical bacilli as they are usually represented to be. For though under many conditions their morphological characters are those of typical bacilli, yet under others they revert to or assume forms indicating their relationship to *Saccharomyces* or a still higher mycelia fungus. In the case of anthrax, the typical bacilli may be represented by oval and spherical bodies, some of which may contain vacuoles, and under conditions (early stages of growth on plates composed of beef bouillon, gelatin 10 per cent., pepton 1 per cent., salt 1 per cent.), the colonies are composed of large spindle-shaped, spherical or oval elements in which vacuolation is frequent. Similar appearances are to be observed in colonies of the thrush fungus. From this it is inferred that while *B. anthracis* is a typical bacillus as a pathogenic microbe, yet in its early stages of growth on gelatin it may assume characters having much resemblance to *Saccharomyces mycoderma* or *Oidium* and thus return temporarily to an atavistic stage in its evolutionary history. With regard to *B. diphtheriæ* the author points out that the club-shaped expansions of one or both ends are not to be regarded as due to involution, for both under natural and artificial conditions where there is active growth these expansions will be found, and have moreover a striking resemblance to the ends of growing hyphæ. Their existence, therefore, is only to be explained by their representing a relationship to a mycelial fungus. In the case of the tubercle bacilli, preparations not unfrequently show threads or filaments composed of unequal elements, some of them being conspicuous for knob-shaped expansions, similar to those of diphtheria. Such appearances occur not only in sputum but in artificial cultivations e.g. glycerin agar after some weeks incubation at 37°. All these preparations behave in the same way as *B. tuberculosis* when treated with appropriate staining reagents; and that they are not involution forms is evident, as the unbranched nature of the

filaments and the existence of lateral bulgings prove that they are in an active condition of growth.

Lysidin.—Ladenburg describes a compound obtained in the state of hydrochloride by heating ethylene diamene hypochloride with sodium acetate. The composition of the freebase is $C_4H_8N_2$ and is termed *lysidin*. The aqueous solutions dissolve uric acid and the application of lysidin in the treatment of diseases arising from the secretion of uric acid is being investigated. Grawitz describes it as a crystalline body of a light red color, readily soluble in water and possesses a peculiar taste. It is administered in doses from 15 to 80 grains daily, dissolved in carbonic acid-water.—*Deutsche med. Wochenschr.*, 1894, 786.

Gaseous Formaldehyde.—R. Cambier and A. Brochet prepare this aldehyde for disinfection in two ways: 1. By the depolymerization of trioxymethylene by heat, and, 2. Direct production by the incomplete combustion of methylic alcohol. Formaldehyde possesses antiseptic properties only when it is in the condition of a gas. On cooling, ordinarily, it is spontaneously polymerized to an inert solid. If it is allowed to cool, in the presence of much air this process does not take place and hence the formaldehyde retains its bactericidal properties. Experiments made at the bacteriological laboratory of Montsouris have enabled the authors to sterilize the ordinary dust of rooms as well as cultivations of various pathogenic micro-organisms.—*Compt. Rend.*, 1894, No. 15.

NOTES HERE AND THERE.

Soda Water.—In Byron's "Don Juan" we find the following in Canto II., 81, 186:

Ring for your valet—bid him quickly bring
Some hock and soda water, then you'll know
A pleasure worthy Xerxes, the great king;
For not the best sherbet sublimed with snow,
Nor the first sparkle of the desert spring,
Nor Burgundy, in all its sunset glow,
After long travel, *ennui*, love or slaughter,
Vie with that draught of hock and soda water.

The Essence of Rose Industry in Turkey.—The *Bulletin du Musée Commercial*, in its issue for September 1st, states that the essence of rose industry in Turkey, which was until quite recently one of the principal resources of Eastern Roumelia and the principality of Bulgaria, has within the last few years shown a decided decline, the falling being the quantities and values of the exports during that period:—1889, 2,767 kilos., valued at 1,542,544 francs; 1890, 3,163 kilos., valued at 1,771,427 francs; 1891, 534 kilos., valued at 317,937 francs; 1892, 439 kilos., valued at 267,379 francs. In 1893 the value of the exports was only 143,185 francs. This decline is due largely to the fact that in France, Germany, and in several other places in Turkey besides Roumelia a development has taken place in the growing of roses, so as to provide to some extent for the requirements of consumption in these places.—*Brit. and Col. Drug.*, 1894, 421.

Alumni Association.

Minutes of the Executive Board meeting held January 9, 1895.

The meeting was called to order at about 8.30 P. M. by the President. There were present Miss K. C. Mahegin and the Messrs. Graeser, Henning, Ehrgott and Hoburg.

On motion, the reading of the Minutes of the last Executive Board meeting was dispensed with.

Reports of Committees:

The Letter-Box Committee reported progress, and that the "box" will be up in a few days.

Motion made and seconded that the Alumni Room Furnishing Committee be discharged with the heartfelt thanks of the association, and that the Secretary notify the chairman of said committee, Mr. Hohenthal, of this action. Motion carried.

The report of the Treasurer was very satisfactory, and was forthwith adopted.

The business manager of the JOURNAL reported it as being in a very flourishing condition, which reassuring report was gladly adopted.

After having duly notified the following gentlemen, they were to-night dropped from membership in the Alumni Association, a motion, which was seconded and carried having been made to that effect, and that the Secretary request the return of their certificates of membership, according to a clause in our Constitution to that effect. These gentlemen are Messrs. George W. Snedeker, A. Zimmerman and A. T. Halsted.

The resignation of W. M. Rheineck was recently received, and since he gave sufficient reason for so doing, his resignation was accepted with regrets.

The resignation of Mr. A. Henning as Business Manager of the JOURNAL was also handed in this evening, and under the existing circumstances it had

to be accepted, with the sincerest regrets of the association.

It was then regularly moved and seconded that the salary of the editor of THE ALUMNI JOURNAL be increased on account of three extra issues of the JOURNAL per annum.

After a very interesting discussion of important business for an hour or so, the meeting came to a pleasant termination.

W. A. HOBURG, Jr., Sec'y.

The following list of names are of persons who have changed their addresses and consequently the Treasurer of THE ALUMNI JOURNAL is unable to supply them with the information that they are entitled to. If these persons or any one knowing of their addresses will communicate with Mr. A. Henning, this end will be attained:

Adam Vogt, 787 8th avenue, city; A. Levy, 125 Grand street, city; G. J. Wolston, Cortland, Cortland Co., N. Y.; H. W. Walp, 536 5th avenue, city; Gustav Katz, Lenox avenue and 125th street, city; Alfred Miller, 537 9th avenue, city; Fred. T. Hartman, 703 3d avenue, Brooklyn, N. Y.; Thos. H. McDonald, Cairo, Ill.; A. J. Van der Bergh, 213 6th avenue, city; C. E. W. Lewin, 106 2d avenue, city; Emil Th. F. Holthusen, 20 Rutger street, city; Emil Buchler, 100 St. Marks Place, city; Frank K. Burr, 821 7th avenue, city; A. W. Moschowitz, 1099 Broadway, city; L. D. Huntoon, Port Oram, N. J.; Chas. E. Stammeler, 172 Varick street, city; Chas. H. Everest, 27 West 34th street, city; Edward Stone, 1501 Broadway, city; Fred. Peiter, 301 3d avenue, city; Major C. Brown, 874 Broadway, city; Louis Hess, Scranton, Pa.; A. Zimmerman, 561 5th avenue, city; Otto C. B. Groin, Denver, Col.; Jacobo Alvarado, Paso del Norte, Mexico; G. S. Badger, 52 East 42d street, city; Frank A. M. Schleiff, 242 East 27th street, city.

“We’ll learn the perfect skill,
The nature of each herb to know,
Which cures and which can kill.”

College Notes.

MARRIED.—Smith Ely Jelliffe, M. D., to Helena Dewey Leeming, both of Brooklyn, by Rev. Dr. Kelsay, of Brooklyn, assisted by Rev. T. LaFleur, of Montreal, Thursday, Dec. 20th, 1894. In the 6th Ave. Baptist Church, Brooklyn, at 8 P. M.

'94 NOTES.

Apropos of the New Year, it is seemingly proper that we should endeavor to surpass our former records by carrying out such resolutions that we may deem proper both for the welfare of ourselves and the gratification of our associates.

At the present time, I think one of the most important resolutions should regard the memory of our Alma Mater. Therefore let me suggest that the bonds of friendship that have hitherto existed, be not cast asunder, but on the contrary, be more tightly strengthened. Let us in the strife and turmoil of commercial life, pause, if but for a moment and think of the pleasant days spent at college, the recollections of which not even time can efface from our memories.

To enable us carry out this resolution, our Alumni Association has extended their characteristic hospitality by inviting us to their monthly lectures, therefore why should we not show our appreciation of their kindly feeling, by taking advantage of the opportunity, and thus not only serving to further make these meetings enthusiastic and successful ones, but also demonstrating to our fraternal friends that sociability is not a lost art among us.

EX-SECY. INHOFF is at present in Colorado seeking the high altitude of the Rocky Mts. as a substitute for the many panaceas, usually recommended for obesity. Last reports were to the effect that the trip was not taken in vain.

Despite the prevailing rain and cold winds, many of our “Gilded Pharmacists” braved the elements in order to have Prof. Haubold give them a few “pointers” on digestion. It is needless to say that they were liberally rewarded, for, who would not enjoy the pleasure of an “Iodine Sandwich with a test tube of genuine pancreatic juice on the side,” handed him, particularly when the latter was the self-sacrifice of a wandering specie of canine.

Our class was represented by Messrs. Race, Burger, Ely, Hutchinson, Struck, Pond, Krueder, Katz, Wurthiman and Stoezer, who did justice to our familiar. Pento! Meta! Boraci!

EX-SEC’Y LINNIG has been advised by his physician to drink no more water as its reaction on his cast iron constitution might result in an incrustation commonly known as Rust.

MOSE KATZ as bright and jovial as ever is still with Messrs. J. N. Hegeman & Co., 3d Ave. and 31st St. He anticipates being present at most if not all of the Alumni lectures this winter.

FRED HILTZ left for Cleveland, Ohio, a few weeks ago. He anticipates entering the Medical University of that city next year; subsequently he will finish in the P. and S. College, this city under the guidance of Harry W. Carter, Ph. D., A. M., of Brooklyn.

JOHN P. WILCOX is located in Plainfield, N. J.

One of our most successful graduates is AUG. W. BRATER, who together with his brother is conducting a cosy pharmacy on Park Ave., cor. 76th St. Brater is as energetic as ever and devotes no little time in making an exquisite window display, which is the admiration of the neighborhood’s fair ones.

ARTHUR BASTEDO is indeed quite a genius, for besides attending to his duties with Caswell & Massy, he has found sufficient time to dissect several times a week at the P. and S. College, which will be an advantage to him

when he commences the study of medicine. Arthur has also joined the Alumni Association and is such an active member that he may be found at all their meetings.

Through the endeavors of J. REMINGTON WOOD (with a little bunch of whiskers on his chin), we hope to have a reunion dinner before commencement. His success on former committees of this kind gives us every confidence of his ability to make such an occasion a success at this time.

THOS. E. DAVIES is hospital steward of the Eighth Battalion, N. G. S. N. Y., and a quite popular one too. At their receptions and drills the Red Cross of his uniform is always conspicuous. He spent two weeks in State camp during the summer, of which his reminiscences are many as well as interesting. Mr. Davies has just met with a severe loss in the death of his Father.

NELSON S. KIRK, PH. G.,
9 E. 59th St.

Senior Class Notes.

D. M. WELLS on returning home one evening found his room in a somewhat disjointed condition. The bed was taken apart, pillows tacked to the wall, and books, clothes, ladies' photos and old suspenders heaped up in artistic fashion on the floor. He thought the place was struck by lightning, but was informed that it was the work of a couple of friends who had called to see him.

The servant girl has a gun loaded. So beware, Cooley.

Wells says home coming is not pleasant when you have to climb through the transom to get into your room.

For the Johnson & Johnson excursion Brown is going to have his whiskers trimmed, Manville is having his voice scoured; Joe is going to wear his new white hat; Gifford is going to have his hair cut so as to disguise himself; Morse and his extra eyes will be there too; Clarey says I am going if my fair one does too.

Thum is going to have his trousers pressed and his hair banded.

Sherman is going to put glucose on his mustache to swap for cold sores.

Cooley says, no, thanks, I have had the grip twice this year: no cold sores in mine.

Dalton is going to try and keep awake during the entire trip.

The things which are troubling the students:

First—Is New Brunswick a prohibition town?

Second—Is there to be any acts between the drinks?

Third—How many slices of ham between New Brunswick sandwiches?

Messrs. Steihener, Scharnibon and Koerber have been appointed by section one a committee to furnish sauer kraut for that section while on the excursion.

All the boys they will be there,
Vanderbeck will comb his hair,
Kneuper will flirt with the ladies sweet,
While Ferguson cries, when do we eat?

Roberts will bring in his tambourine,
Watling will sing when he is not seen;
Bricks will be placed in easy reach
In case he is discovered making such a breach.

Flick will make a mash I am sure,
While on that plaster hunting tour:
For who could resist such charming eyes,
When on them Flicky only tries.

Boenke will give a song and dance,
McClellan will go quietly off in a trance,
The Heffley boys will spin some jokes,
Which are rivals in age of the mighty Oaks.

MR. H. E. COOLEY, who had a slight attack of the grip, is around again to the rejoicing of his many friends.

The action of the class in requiring its candidates for Valedictorian to enter a speaking contest to determine their fitness, meets with the general approval of all its members.

MANVILLE admitted that he was Hazy. How about replacing that H with L.

AN INSTRUCTIVE TRIP.

A very entertaining and instructive visit was made by a number of students of the senior class, on Saturday, Jan. 12th, to the Mineral Water Works of Dr. Carl H. Schultz.

The trip was arranged by the Pharmaceutical Club, of 37th East 19th St., represented by Mr. T. B. Dean, its corresponding secretary, which seems to be especially active as regards our interest and welfare and extends to us the fostering care of a parental guardian. It is due to this club's hospitality and magnanimity that our Glee Club has thrived so wonderfully.

Mr. Dean kindly introduced us to Mr. Louis Waefelaer, M. E., the assistant chemist of the works (Dr. A. P. Hallock, Ph. D., the chief chemist and Dr. Schultz being away at the time), and Mr. Paul Dimmer, the foreman. These gentlemen, starting at the beginning of the works where the croton water enters by five different mains, and followed the course of the water through each step of the process, whereby the water was filtered, then heated to destroy organic as well as to drive off decomposing and volatile organic matter as well as other impurities and the filtered water there distilled by the most practical and complete apparatus conceivable; then the water was repeatedly subjected to tests, for various impurities, in their admirably equipped chemical laboratory, which is also supplied with a room specially devoted to bacteriological work, and a dark room for spectrum analysis and photographic investigation. Here also are prepared the solutions used in making the various mineral waters and where the finished product of the factory is brought before being sent out in order to be tested and to make doubly certain that it agrees with the label bearing the analysis of contents, which is placed on each siphon of water sent out. Here also we quenched our thirst with the products of the stills of this as well as with the products of the stills of other factories.

The carbonic acid gas used in charging the waters also passes after generation through a set of coolers, mashers and purifiers, to completely remove all impurities, and is stored till required for charging.

The whole establishment, embracing nineteen different departments, employs over 250 men and 100 horses; the fountain, bottle and siphon filling department has a capacity of 50,000 siphons or 10,000 gallons per day. The elaborate machinery of the works is mainly the invention of the proprietor, his deceased son and staff; not the least important among which is the invention of Mr. Paul Dimmer.

Mr. Louis Waefelaer, the assistant chemist, is a young mechanical engineer of high standing and has sole charge of the mechanical department. Every department is scrupulously clean and neat, and the employees think Mr. Schultz is one of the best and most liberal men to work for, for he spares no expense in investigations and experiments calculated to improve the accuracy and purity of the products of his works, and the safeguards against accident to employees are both numerous and well devised. Several other parties will be formed, from the senior class, during the course of the term and will visit and be shown the workings of this "model establishment."

CLASS REPORTERS.

Junior Notes.

IN MEMORIAM.

B. C. MEANEY, entered into rest, Sunday, January 6, 1895, in the 22d year of his age. This brief announcement reminds us of the loss and sorrow to so many near relatives and friends, that after the few weeks that have elapsed since their hearts were wrung with grief. We venture to say something of him whose earthly sojourn is ended.

Possessed of a genial happy temperament, a character so manly, conservative and refined, that professors as well as students rendered to him an involuntary tribute of respect. In the three months that the junior class has been organized, few students have become better known or more popular than Mr. Meaney.

Just before the college closed for the Christmas vacation, he said to a friend, "I think this will be the happiest Christmas I have ever had," and now who that knew him can doubt that this strange prophecy has been fulfilled.

J. Y. C.

CLASS MEETING.

The meeting was called on Tuesday, January 8, 1895, by the death of our classmate, Mr. B. C. Meaney. A motion was made that we send flowers to his late home, which was amended so as to include the drawing up of resolutions of condolence, and sending a copy of them to his parents. Carried.

The meeting then adjourned.

F. H. FINLEY, Sec.

Before vacation it was rumored that our friend and professor, Dr. Jelliffe, was about to become a benedict, and as the rumor has become verified, we, the Class of '96, send to him our hearty congratulations and best wishes for a long and happy life.

There is one thing the Juniors should pay more attention to, that is class meetings. If each one who could would come, the difference would quickly be seen. Try it.

The Juniors in pharmacognosy commenced work with the compound microscope at the beginning of the term.

On exhibition every Tuesday afternoon, from 4.30 to 5, in Quiz, T.'s hand.

We are sorry to hear our friend and classmate, Mr. Quickburger, has been hurt, and hope it is nothing serious. He was thrown from a cable car against a post on Tuesday, and was picked up insensible. The car was just making the turn, which it does in a rapid manner, and it is supposed he had no hold.

A great many cases of mustaches have broken out among the Juniors. In most cases, however, it is only a light attack, and not at all serious.

They say the back part of the Botany Quiz room was very warm the other day; in fact, some of the boys were nearly roasted.

Did I hand in that joke I heard in Quiz the other day? If not, why not? It would have helped to make the page interesting this month. Two weeks no college. Reporter with one week. He will do the best he can, but every little helps.

Remember, this page is for the Class, not individuals, and every time you help make the Junior page interesting you are doing the Class a favor as well as the reporter.

All communications for Junior notes should be addressed to

J. Y. CANTWELL,
261 West 42d street.

MEDICINE AND PHARMACY.

BY N. H. MARTIN, F. L. S., K. R. M. S., President of the British
Pharmaceutical Conference.

(Continued, from December issue)

Doctor's dispensing is stated by many to be one of the chief if not the chief cause of the ills from which pharmacy is a sufferer, and demands in more or less dignified terms are made that this iniquity shall cease. I make no apology for the existence of this condition of things. Theoretically it is undoubtedly better that dispensing shall be done by the pharmacist, and prescribing by the medical man, but when we pharmacists claim this as a right, and accuse medicine of unjustly usurping our functions, it is well for us to remind ourselves that medical men, although they may not now as frequently as of old take the degree of L. S. A., are the direct and legitimate successors of the old apothecary and that the dispensing of medicine was their legitimate function. So much was this the case that there being a doubt as to whether it was traversed by our own Act of 1868, the short Act of 1869 was passed to preserve the right. Then again it is deep rooted in the habits of the English people to expect the doctor to supply the medicine he has prescribed, and any change can only come about by the slow process of educating the patients and by the exhibition of good will and feeling between medicine and pharmacy. Before it can happen universally there is no doubt that pharmacy must have acquired such a professional standing and education as will enable it to perform its delicate and confidential function with the tact and reserve which is the outcome of prolonged training. The mistake (a very common one) which pharmacy is making, is that it wants the reward before it has made the effort and suitably equipped itself for the service. I exhort the pharmacist of the future to be unremitting in his efforts to raise himself and his calling to a professional status, and then I predict for him that in the natural course the dispensing of medicines will come to him.

Chemist's prescribing is quite as loudly complained of by the doctors, and when I read some of the letters and comments which appear in the

medical journals I am almost tempted to fear that for once medicine is thinking more of its share of the pecuniary reward, than caring for suffering humanity. There is, however, I am sorry to say, a great deal too much prescribing by chemists, and some of it is of a most reprehensible kind. I know a case where a chemist treated a man suffering from rodent ulcer of the face for two years, all the time buoying the man up with the hope that it was getting better, and that he would cure it, until the face was so bad, and the ulcer had spread to such an extent that when it came under the notice of the surgeon nothing could be done for the patient. If that chemist had met the man upon the highway, and robbed him, he would have been liable to imprisonment, but having got the man into his shop he not only robbed him of his money, but he rendered it impossible for the man ever again to be restored to health. For the dishonor which such men bring upon pharmacy, and for the irreparable injury which they inflict upon suffering humanity I should like to give them several years of penal servitude. But there are innumerable small accidents, and little ailments to which humanity is liable, which quite legitimately come within the province of pharmacy to treat, and the pharmacist, if he is wise, is a much safer man to treat these than the clergy and the laity, who are ever ready to prescribe for each other upon any and all occasions. The best and wisest exponents of medicine admit this right on the part of pharmacy, and welcome the service which is rendered by it to sufferers. Pharmacy may make some mistakes, but I know it frequently sends patients to medicine long before they or their friends would think seriously enough of the case to do so.

There should be no rivalries or jealousies between medicine and pharmacy, and the better qualified each of these may be to exercise its own share of the duties devolving upon both, the more will each of them respect the rights and the work of the other.

Before I conclude, one word on the principle upon which remuneration should be based. This is a question of the utmost importance to the English public, as well as to the pharmacists. John Ruskin says, "You do not pay judges large salaries because the same amount of work could not be purchased for a smaller sum, but that you may give them enough to render them superior to the temptation of selling justice." We cannot err in applying this principle to pharmacy, and deciding that the dispensing chemist must be paid at a rate of remuneration which will enable him to get

his living honestly and openly, and render him superior to the temptation to increase his profit and his income by tampering, in ever so small a degree, with the quality of the drugs he uses, and with the health, and may be the lives, of dear ones, and of men important to the community. His remuneration should also enable him to devote sufficient time and care to every detail of his responsible work, and eliminate a very real source of danger which is unavoidable if the haste and the bustle of trade methods are adopted by pharmacy.

The Conference has entered upon the fourth decade of its existence, and, possibly, I should have made a better and wiser choice if I had addressed you upon its past achievements, and its future prospects, but the other matters upon which I have touched seemed to me of greater importance. Let me say, however, briefly, that I think the record of this Conference has been eminently an honorable one, and that it has fulfilled, in a high degree, the functions for which it was called into existence. The story is written in the Year Books, and another phase of it is engraved in the hearts and memories of many of us who have been members almost from the beginning, and who have attended a large number of its meetings. It has added to our knowledge, enlarged our experience, and broadened our intellectual grasp of pharmacy; and last, but not least, it has been the means of bringing together, introducing to each other, and cementing friendships between men who practice a common avocation in districts as wide apart as Inverness and Cornwall. In this latter function the excursion on the last day has played no inconsiderable part. Amongst the critics of the Conference there are some persons who affect to sneer at the excursion as if it were sheer frivolity, and was at variance with the avowed scientific objects of the Conference. I beg to differ, and to claim for the excursion day a very high place in the work of the Conference. It affords the opportunity, as no other arrangement could do so well, for men to meet; and I am quite sure that my own experience is by no means singular when I tell you that many, very many, of the best friends I have in pharmacy were first known to me through the opportunity of one of the Conference excursions; and further I could not exaggerate to you the benefit which I have received from the numerous conversations and informal discussions which always takes place on these days. But it is with societies, as with individuals, they tend to decay, and already, more than once we have the alarm: the Conference is on its last legs! I do not believe it, as I feel sure it fulfils a purpose in the realm

of pharmacy which is too important for the Conference to be left to decay, and if we neglect the trust which has been handed down to us, our successors will revive it. I would ask every member of the Conference to get, at least, one other member to join, and I do not think he can use a stronger argument, than that, apart from the opportunity of attending and taking part in this annual scientific gathering of pharmacy, the Year Book, which he will receive, is worth many times the subscription. The Year Book of Pharmacy should find a place on the desk of every chemist and druggist in this land. In it he will find abstracts of papers from a larger number of sources than he can possibly consult for himself, and many of these papers may be of great value to him.

There is no occasion to disguise the fact that we do not get as many or possibly as good papers sent to the Conference as we should like, but when we consider the needs of a weekly press and the number of small societies which absorb in the aggregate a large number of papers, our experience need cause us neither surprise nor alarm. I should like, however, to ask many of those who are doing original work and writing papers in connection with pharmacy to consider whether there is any place so suitable for them to be read as at these meetings.

The authors may feel certain of a larger audience to listen to their papers and a far more capable set of men to discuss them than can be found at any other time or place. In provincial towns the papers are read to a few local men, and the discussion is taken part in by fewer still, and even at the monthly meetings at Bloomsbury Square the discussions have a great tendency to fall into the hands of very few men. However capable these men may be, they cannot possibly have the wide and varied experience of the aggregate of the men who attend this Conference. I would, therefore, venture to urge thoughtful pharmacists to contribute papers to this Conference, and I should like them to come in such numbers that we may be compelled to add another day or two to our meeting.

I mentioned just now the friends whom we have met at these Conference meetings, and before I close I must briefly allude to those we have lost. The first name that will occur to you, I am sure, is that of our genial botanist, the late Professor Bentley, who was president at Nottingham in 1866 and Dundee in 1867. Many of us knew him first and best at Bloomsbury Square as our dear and honored teacher, but to many others the Conference must

have been the means of their meeting him, and by all was he respected and beloved. He reached a good ripe age, and of him it might be said—as of many other men who have lived and been true to themselves and their calling—“He has done his work well and earned his rest.” The next, an even greater loss to us as a Conference, because of his younger age and the promise there was in him of greater achievements for pharmacy, is our late treasurer, Mr. R. H. Davies, I, with many others, made his acquaintance through this Conference, and I feel, as I am sure many of you do, that I have lost a personal friend with whom intimacy would have ripened year by year into stronger bonds.

OFFICINAL OR OFFICIAL.

In the *Pharmacentische Rundschau* for January, 1895, is found an interesting discussion on the use of the words officinal and official by Theodore Husemann, of Göttingen, and Charles Rice, of New York. It would be interesting to our readers to give the views of both of these well-known writers in full. At present, however, we reprint in full the views of Dr. Rice:

“In compliance with a request by the editor of this journal, the writer presents a few facts, as well as his personal views, regarding the use of the words “official” and “officinal” when applied to drugs and medicinal preparations.”

It should be stated at the outset that the writer accepts the ordinary derivation of the two words, and the meanings assigned to them in accordance with their origin. Nor does he deny that it has been customary, up to within a few decades, to apply the English word “officinal” quite generally in the sense of “pharmacopœial.” Yet, within the memory of most readers of the *Rundschau*, voices arose in favor of a change, the word “official” being proposed to replace “officinal” in the special sense of “pharmacopœial.” It is evident that some cause arose which produced the feeling that such a change was necessary and the cause is not far to seek. In those countries in which the exercise of pharmacy is under the control of the government, and where the stock of a pharmacist, so far as it is used in physicians’ prescriptions, contains comparatively few remedies besides those directed by the Pharmacopœia, the two meanings of the word “officinal,” viz: 1, the original one “pertaining to an ‘officina;’ pertaining to or kept in a drug store,” and, 2, the more modern one, “pharmacopœial; authoritative,” practically cover each other. This is particularly the case in Germany, where the word “officinell,” and in France, where “officinal” is in general use in the second sense mentioned above. It is different in this country, where the pharmacist is compelled to carry a large stock of non-pharmacopœial preparations, many of which are prescribed by physicians.

The two meanings of the word “officinal” have two widely differing boundaries. They may be likened to two concentric circles. In the first mentioned sense (“kept in a drug store”) the word occupies the area of the larger circle; in the second sense (“pharmacopœial”) usually that of the inner, smaller circle. In some parts of this country the inner circle—to continue the simile—is much smaller in proportion to the outer than in others. In some it may attain an area of perhaps three-fourths or four-fifths of the larger; in others it may even outgrow the former outer circle. Only in rare cases will the peripheries of the two circles coincide. Since the two meanings long ago ceased to cover each other, the necessity arose to use different words to express the two different meanings, and it was therefore, proposed to employ the closely related word “official” in the sense of “pharmacopœial,” and to use the word “officinal” only in the general sense “kept in a drug store,” which is, indeed, in accordance with its original meaning and origin. Those who object to the use of “official” in the sense of “pharmacopœial” say that *officialis* means “governmental; pertaining to an office or official, etc.” That it is, therefore, correct to say, for instance: “The official preparations for the reception of the President are completed,” but incorrect to say: “He made all the official preparations in his own laboratory.” There is, however, no danger of any misunderstanding in these two sentences, indeed, much less danger than would be “officinal.”

Professor Husemann, in his letter, brings within the space of his discussion the terms “*medicamenta magistralia*,” and “*formulæ magistrates*.” He shows, himself, that while the word *officinalis*^[1] was, in more recent times, applied to drugs and preparations of an authoritative character or origin, it was formerly used in its broader sense “what is at any time to be had in a drug store,” in which sense it was the opposite of *magistralis* (magistral, or magisterial), or that which is not kept ready made, but has to be prepared or compounded extemporaneously. It will be noticed that there is a much better logical correspondence between the terms

Medicamenta magistralia = medicines whose composition is fixed or prescribed by the *magister* (a person), that is the attending physician, and

Medicamenta officialia = medicines whose composition is fixed or prescribed by an *official* (a person), that is the Committee of Revision as a body—

than there would be between the former and *medicamenta officinalia*, which term refers to the *shop* and not to the *person* of authority.

As to the word “unofficinal,” this means properly “not pertaining to, not kept by or dealt in by a pharmacist.” If used in this strictly literal sense, however, its scope or applicability will become more and more contracted in the course of time, as it may eventually become difficult to mention articles to which the word may justly apply. It should be abandoned altogether. “Unofficial” much better expresses the idea sought to be conveyed by it. A few examples will show the use and meaning of the several words: Fleming’s tincture of Aconite is not official (or “Unofficial;” not “unofficinal,”) but it is officinal. Tinctura Opii Deodorati is official, and ought to be everywhere officinal.

Concerning the right of any person, or body of men, to coin a new word, or to use one already in existence, for the purpose of expressing a new idea, or removing an ambiguity, there can be no question, provided only that the selected word be appropriate and in harmony with the genius of the language. Of course, its acceptance by the public at large, or by the profession, for the use or benefit of which it was coined or selected, cannot be enforced. Yet, if it is found to answer its purpose, and if its superiority over the term formerly used in place of it is recognized, it will gradually and surely come into general use.

The judgment of the writer is that the employment of the word “official” in the sense of “pharmacopœial” is justifiable on linguistic grounds, and that it is, moreover, fully justified by the condition of pharmacy in this country, where a clear distinction between “all sorts of medicines,” and “pharmacopœial medicines” has become necessary. Of course, the Committee of Revision,” which hoped to settle the controversy by an “official” vote, according to which the word “official” was hereafter to be used in place of “officinal,” when applied to pharmacopœial preparations or directions (see U. S. Pharm., 1890, p. xxxvi.), did not mean thereby to encroach upon the ordinary meaning of the word, which appears, for instance, on the title page of the Pharmacopœia in the sentence: “Official from January 1, 1890.”

[1] Professor Husemann did not find this word in *Du Cange’s Glossarium Mediæ et Infimæ Latinitatis*. It is, however, contained in the latest edition (by

Favre; Niort 1883-87), Vol. VI. p. 37.

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